

RUDRA CHAVDA

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EDUCATION

University of California, Santa Barbara	2026 - 2028
University of California, Santa Cruz	2024 - 2026
Bachelor of Science, Computer Engineering	GPA: 3.95
– Honors: 6x Dean's List – Coursework: Data Structures and Algorithms, C Programming, Computer Architecture, Operating Systems, Software Engineering.	

TECHNICAL SKILLS

Languages	Python, C, C++, Java, TypeScript, JavaScript, SQL, Assembly
Frameworks	Vue.js, React, Next.js, Node.js, Py4Web, Tailwind CSS, Angular
Tools	Git, Nautilus (Cluster), CARLA, Docker, GDAL, VS Code, Google Colab, Machine Learning
Concepts	Reinforcement Learning, Data Structures, Algorithms, System Design

EXPERIENCE

AIEA Lab (UCSC)	Dec 2025 – Present
<i>Undergraduate Researcher</i>	
– Wrote and benchmarked Reinforcement Learning (RL) algorithms such as DQN, A3C, and Stable Baselines in the Car Racing environment to evaluate autonomous vehicle safety as part of the Robustifying AV's project. – Deploying the CARLA simulator in a headless environment on the Nautilus hyper-cluster (Kubernetes). – Analyzing edge-case failures by visualizing agent behaviors and reward functions post-training.	

CA-Futures (UCSC EEB & Baskin Engineering)	May 2025 – Sep 2025
<i>Software Developer</i>	
– Developed a full-stack conservation modeling application using Vue.js (frontend) and Py4Web (backend). – Integrated geospatial analysis tools (GDAL, Ecoscape) to track species connectivity and visualize ecological data. – Optimized simulation parameters to process large-scale biological datasets for the Ecology Lab.	

Opennote (YC S25)	Jul 2025 – Present
<i>Campus Growth Lead</i>	
– Co-organized UCSC Cruzhacks in collaboration with Opennote where we gave out \$500 worth of prizes. – Spearheaded campus growth strategy, increasing user engagement by 30% and acquiring 50+ active users.	

Fassur	June 2023 – Oct 2023
<i>Front-End Engineering Intern</i>	
– Architected a secure, interactive web application using React.js and Tailwind CSS. – Designed a responsive front-end framework to improve data visualization for client dashboards.	

PROJECTS

Better Notes TypeScript, Tailwind CSS, Vercel	Spring 2025
– Engineered a full-stack note-taking platform with a custom markdown rendering engine for student study workflows. – Secured 100+ active users within 3 months of deployment to Vercel. – Live Demo: betternotes-two.vercel.app	
Div Explorer Project Machine Learning, Python	Fall 2025
– Developed actionable data labeling pipelines to detect duplicate discoveries and high-overlap patterns. – Implemented categorization metrics (interpretability, coverage, anomalies) to improve ML model fairness.	
Personal Portfolio Next.js, Three.js, GSAP	Winter 2024
– Built a high-performance portfolio site featuring 3D interactive elements using Three.js and GSAP animations.	

LEADERSHIP

Eagle Scout Project	Summer 2024
<i>Project Lead, Troop 537</i>	
• Directed a team of 20+ Scouts in the construction of redwood structures for Sycamore Grove Park (Livermore).	